

9 Tropical Forest Hydrology

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Abstract

The consequences of land-use and climate change in tropical forest regions are considered among the greatest environmental concerns for the survival of the human population today and include increases in stream flow and flood frequency, loss of soils, changes in nutrient fluxes, warmer and drier climates and concomitant hydrological ecosystem services at the local scale. To tackle all aspects of these hydrological problems, we have broad recognition and tacit acceptance that progress on these complex issues benefits from fundamental pieces of knowledge on all of the hydrological components in various types of tropical forests and synthesizing them in terms of the global implications. Therefore, this chapter's main objective is to provide a pan-tropical perspective on tropical forest hydrology.

9.1 Introduction

The objective of this chapter is to provide the state of knowledge on the characteristics of hydrological components of precipitation, evapotranspiration and streamflow generation in tropical forest regions by a review of tropical literature. This uses a pan-tropical perspective partly via the use of pan-tropical maps of the hydrological components. We will suggest research needs and strategies for studying tropical forest hydrology in a dramatically changing world.

The tropical domain has the largest proportion of the world's forests (45%; FAO, 2020). One may imagine that the tropical forest regions are generally characterized by higher incident radiative energy, a constantly higher temperature and a large amount of ecosystem water resources, enough to support their higher

primary productivity and active water cycling. In reality, tropical forests play a significant role in the global carbon budget (e.g. Beer *et al.*, 2010; Pan *et al.*, 2011) and are a significant source of global hydrological fluxes, profoundly influencing both global and regional climates (Avissar and Werth, 2005; Spracklen *et al.*, 2012; Poveda *et al.*, 2014; Kanamori *et al.*, 2018).

The tropical forest regions can be roughly classified into two types based on seasonal variations in precipitation: tropical evergreen rainforest with a 'rainforest climate' and tropical seasonal forest with a 'monsoon climate' (Kumagai *et al.*, 2009). Each region can be classified into more climate and forest types (Walsh, 1996; Tanaka *et al.*, 2008). Tropical evergreen rainforests exist only in regions of ample water resources (Kumagai *et al.*, 2005; Kumagai and Porporato, 2012b). Tropical seasonal forests have to cope with seasonal

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droughts (Vourlitis *et al.*, 2008; Miyazawa *et al.*, 2014; Kumagai *et al.*, 2015). As the existence of these tropical forests achieves a delicate balance in terms of their ecosystem water resources, hydrological change in either region could result in significant impacts on ecological patterns and processes (Malhi *et al.*, 2009; Phillips *et al.*, 2009; Kumagai and Porporato, 2012a), in turn affecting feedbacks to the atmosphere (Meir *et al.*, 2006; Bonan, 2008; Kumagai *et al.*, 2013; van der Ent *et al.*, 2014; Takahashi *et al.*, 2017). Some studies have shown that a marked change in regional climate will occur first in the tropics, making tropical forest ecosystems particularly vulnerable in the future (Mora *et al.*, 2013).

Humans have been modifying tropical forest land cover for food and energy production and for developing tropical countries. Such modifications (i.e. land-use change) are being combined with climate change and should be anticipated to impact the regional hydro-climate as well as the local freshwater resources (Bruijnzeel, 1990; Giambelluca, 2002; van der Ent *et al.*, 2010; Wohl *et al.*, 2012; Kumagai *et al.*, 2013; Takahashi *et al.*, 2017). What distinguishes the current modifications is that the entire hydro-climate is now subject to these modifications. The consequences of land-use and climate change in tropical forest regions are considered among the greatest environmental concerns for the survival of the human population today and include changes in streamflow and flood frequency, loss of productive soils, changes in nutrient fluxes, a warmer and drier climate, and concomitant changes in hydrological ecosystem services at the local scale (Bruijnzeel, 2004; Chappell *et al.*, 2004; Krusche *et al.*, 2011; Lawrence and Vandecar, 2015; Gómez *et al.*, 2023). Progress on these complex issues benefits from fundamental knowledge of all of the hydrological components in various types of tropical forests and synthesizing them globally (Bruijnzeel, 2004).

9.2 Pan-Tropical Climatic Regime and Forest Type

Pan-tropic maps of precipitation characteristics are derived from the NOAA's PRECipitation REConstruction over Land (PREC/L) long-term

gridded precipitation data set (Fig. 9.1) (Chen *et al.*, 2002). High radiative energy received around the equator generates the updraft of hot and humid air. Convergence of this hot and humid air (i.e. the low atmospheric pressure system) results in the belt of precipitation (Fig. 9.1a) associated with the Intertropical Convergence Zone (ITCZ) around the equator. Updraft in the ITCZ moves both northward and southward in the tropopause and descends around the Tropic of Cancer and the Tropic of Capricorn. Around both tropics (the so-called 'horse latitudes'), hot and dry air divergence typically creates arid areas and deserts (Fig. 9.1a). This circulation of air masses ascending around the equator and descending around the tropics is called the Hadley circulation.

According to the Köppen climate classification, tropical climates are defined as conditions in which the air temperatures do not fall below 18°C in the coolest month, where the annual mean precipitation (mm) is more than the Köppen arid boundary ($P_k = 20 \times (T_y + x)$) measured in mm, where T_y is annual mean air temperature (°C) and x is the index representing the precipitation patterns such as 0, 7 and 14 for dry summer and wet winter, wet throughout the year, and dry winter and wet summer, respectively). These climates are further subdivided into rainforest (Af), monsoon (Am) and savannah (Aw) classes. The rainforest climate occurs in the zone where all 12 months have a mean precipitation of at least 60 mm. The monsoon and savannah climates have a mean precipitation of less than 60 mm in the least precipitation month but have the least monthly precipitation, more and less than $100 - 0.04 \times (\text{annual mean precipitation in mm})$, respectively.

As Feng *et al.* (2013) suggested, the areas with the largest amount of precipitation always exist in the regions with the most aseasonal precipitation patterns but not vice versa (Fig. 9.1). The rainforest climate is characterized by both a large annual amount of precipitation (Fig. 9.1a) and little or negligible seasonal variation in precipitation (Fig. 9.1b), which are caused by a stationary low atmospheric pressure system around the equator. Tropical evergreen forests can exist in this region because of their constant water requirements throughout the year. High rainfall intensity of short duration is distinctive in the rainforest climate, and in both diurnal

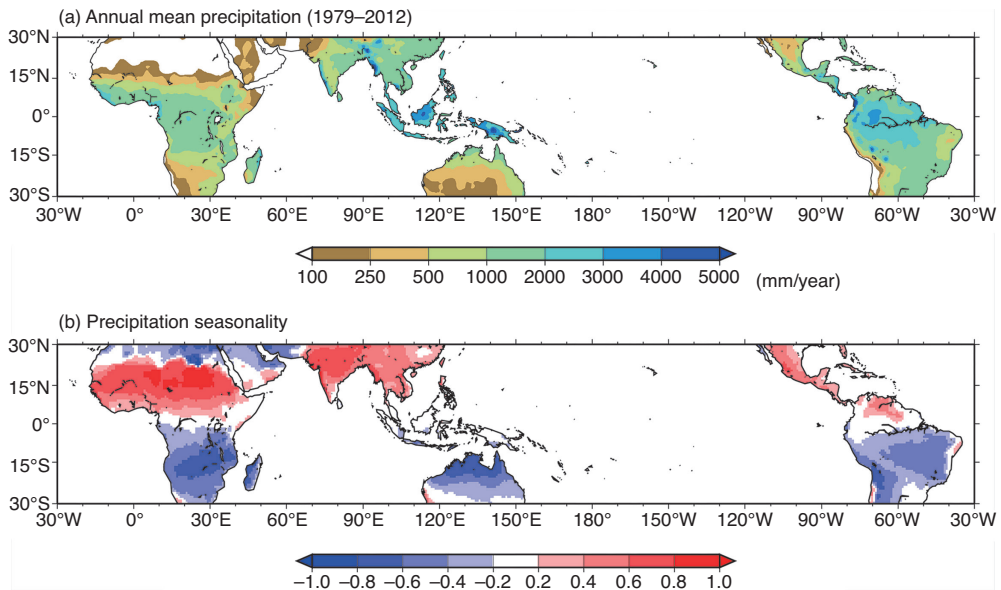


Fig. 9.1. Pan-tropic maps of (a) annual mean precipitation and (b) precipitation seasonality represented by the ratio of difference in precipitation between June–August and December–February to annual precipitation (constructed using NOAA's PRECipitation REConstruction over Land (PREC/L) with values averaged over 1979–2012).

and seasonal variations, there is strongly spatial heterogeneous rainfall distribution over a scale of a single to thousands of square kilometers of tropical rainforests (Krusche *et al.*, 2011; Kumagai and Kume, 2012; Kanamori *et al.*, 2013).

By contrast, the monsoon and the savannah climates are characterized by their degree of seasonality in precipitation caused primarily by seasonal fluctuations of the ITCZ. It should be noted that annual precipitation decreases with the increased seasonality of precipitation but not vice versa (see Feng *et al.*, 2013; Fig. 9.1). There is a less intense dry season and larger precipitation in the monsoon climate than in the savannah climate. Tropical seasonal forests can exist in monsoon areas with climate adaptations that cope with water stress during the dry season (e.g. full- or semi-deciduous traits, seasonal variations in stomatal behavior). This, in turn, means that tropical seasonal forests can vary from evergreen to deciduous forests corresponding to the local precipitation regimes, for example influenced by altitude and topography (Tanaka *et al.*, 2008).

The tropics are alternatively divided into three regions based on biotic and geological history (US Department of Energy, 2012): (i) the neo-tropical ecozone (NEO) of South America, Central America and the Caribbean; (ii) the Afro-tropical ecozone (AFR) of sub-Saharan Africa; and (iii) the Indo–Malay–Australasia tropical ecozone (IMA) including regions of India, South-east Asia, southern China, New Guinea and northern Australia. The forests of the Amazon Basin in the neotropics represent the single largest block of intact tropical forests, occupying around 40% of forest biomass in the tropics (Saatchi *et al.*, 2011). South and Central American and Caribbean forests often experience hurricane-induced disturbances (Boose *et al.*, 1994, 2004; Negrón-Juárez *et al.*, 2010). Among the Afro-tropical forests, those of the Congo Basin represent the second-largest intact block of tropical forests (Pan *et al.*, 2011). Central African forests do not experience tropical cyclones but are still subject to large storm events and suffer from severe drought and special wet periods, particularly during the El Niño and the La Niña phenomena (US Department of Energy,

2012). While the dominant plant family in neo-tropical and Afro-tropical forests is Leguminosae, Dipterocarpaceae dominate within South-east Asian forests. Dipterocarps are among the most valuable timber species globally, leading to intense logging pressure within the IMA tropical forests. Indeed, over the late half of the 20th century, more timber was exported from Borneo Island in this region than from neo-tropical and Afro-tropical regions combined (Curran *et al.*, 2004). Although South-east Asian tropical forests represent only about 11% of the world's tropical forests in terms of area, they had the highest relative deforestation rate around 2000 (e.g. Canadell *et al.*, 2007; Pan *et al.*, 2011). An analysis of climatic trends in global tropical rainforest regions over the period 1960–1998 showed that precipitation declined more significantly in South-east Asia than in Amazonia and that the El Niño–Southern Oscillation (ENSO) is a particularly important driver of drought in South-east Asia (Malhi and Wright, 2004).

9.3 Pan-Tropical Evapotranspiration

Evapotranspiration from the forest canopy can be simply partitioned into wet canopy evaporation or canopy interception, and dry canopy evaporation or transpiration. Kume *et al.* (2011) reviewed tropical forest canopy interception ratios (i.e. wet canopy evaporation divided by gross precipitation) from the data of 40 tropical forests, including both rainforests and seasonal forests, and reported that most ratios measured in tropical forests were in the range of 10–20% with a mean of 17%. In contrast, the reported transpiration rates in tropical forests had large temporal and spatial variation in the value of 2.3–4.6 mm/day (e.g. Shuttleworth *et al.*, 1984; Roberts *et al.*, 1993; Cienciala *et al.*, 2000; Kumagai *et al.*, 2004). Bruijnzeel (1990) suggested that in humid tropical forests, the average annual transpiration was 1045 mm (range 885–1285 mm). The resultant variation in the annual tropical forest evapotranspiration range was from around 1000 to 1500 mm and the ratios of annual evapotranspiration to annual precipitation ranged mostly between 30% and 90% (Kume *et al.*, 2011). Kume *et al.* (2011) found that, apparently, in tropical forests, when

the annual precipitation was less than 2000 mm, evapotranspiration increased with increases in precipitation, and when precipitation exceeded 2000 mm, evapotranspiration reached a plateau of about 1500 mm.

There are more complex situations controlling the evapotranspiration of tropical forests. Trees in the tropical monsoon climatic regions tend to cope with seasonal drought by stomatal control, leaf fall and root water uptake at deeper soil depths (Igarashi *et al.*, 2015). Many pristine tropical forests have been cleared for conversion to rubber trees and oil palm plantations (Carlson *et al.*, 2012; Fox *et al.*, 2012). The water use of such plantations is comparable to or exceeds that of the original vegetation (Guardiola-Claramonte *et al.*, 2010; Kumagai *et al.*, 2015; Giambelluca *et al.*, 2016). Significant portions of the cleared tropical land quickly revert to secondary vegetation and, in terms of hydrological characteristics, increasingly resemble the original forest over time (Giambelluca, 2002). Giambelluca *et al.* (2003) pointed out that evapotranspiration from fragmented forests tends to be enhanced by conditions in surrounding clearings. Bruijnzeel *et al.* (2011) claimed the need to pay special attention to the hydrometeorology in tropical montane cloud forests, which are fog- and cloud-affected, and thus small and large amounts of the transpiration and canopy interception are caused due to the small water vapor deficit and the cloudwater, respectively.

Although it is complex to generalize on evapotranspiration, we can compare evapotranspiration from tropical forests with that of other regions. Annual evapotranspiration (*ET*) from ground observations (e.g. watershed water balance, micrometeorological measurements and soil–water balance) can be classified by annual mean temperature (*T*) and represented as a function of latitude (Fig. 9.2, adapted from Komatsu *et al.*, 2012). Despite a large variation in *ET* at each latitude (~500 mm), which might be caused by altitude, forest type, local climate of the forest sites and differences in hydrological observation methods, there is a strong linear relationship between latitude and *ET*. In humid climates of the ITCZ, the high precipitation (*P*) is expected to cause higher *ET* (Fig. 9.2). Note that there are large variations in *ET* in the tropical forest regions, but many can be explained by the

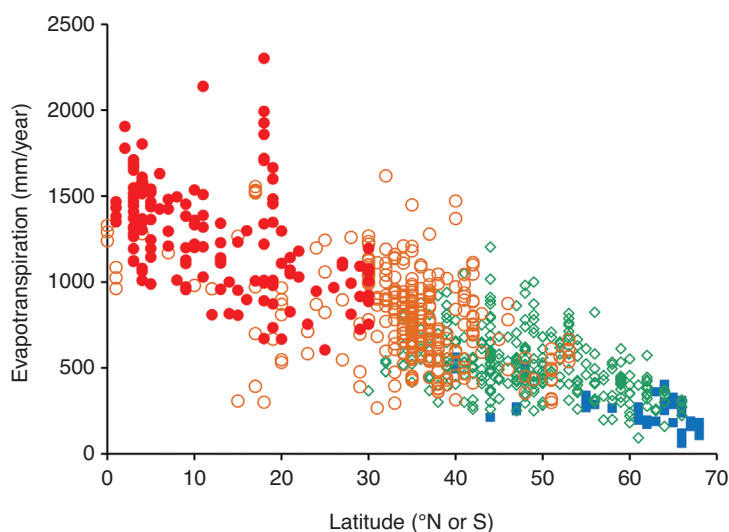


Fig. 9.2. Annual evapotranspiration as a function of latitude, classified by the annual mean temperatures (T , °C): closed circles, $T > 20$; open circles, $10 < T < 20$; open diamonds, $0 < T < 10$; closed squares, $T < 0$. Data on evapotranspiration, latitude and temperature taken from Komatsu *et al.* (2012).

difference in T , probably because of the altitude effects.

From this relationship between annual P and T , Komatsu *et al.* (2012) modified the Zhang *et al.* (2001) evapotranspiration model where:

$$ET = P \frac{1 + w \frac{ET_0}{P}}{1 + w \frac{ET_0}{P} + \frac{P}{ET_0}} \quad (9.1)$$

in which

$$ET_0 = 0.488T^2 + 27.5T + 412 \quad (9.2)$$

where w is a coefficient representing plant water availability ($= 2.0$). ET_0 is the potential evaporation, which was defined as a constant value calculated by the Priestley and Taylor (1972) equation in Zhang *et al.* (2001) but was modified by Komatsu *et al.* (2012) so that it can appropriately describe the temperature effect. Equations 9.1 and 9.2 successfully reproduce ET as a function of latitude (Fig. 9.2). In the tropical rainforest and monsoon climates, it suggests that higher ET eventuates through the conditions of a plentiful P and higher T .

Theoretical (Eqns. 9.1 and 9.2) and observed relationships between annual P and ET are investigated in Fig. 9.3a. The shallower slope in the relation between ET and P in the neotropics (Fig. 9.3a, top) and IMA tropics (Fig. 9.3a,

bottom) compared with the steeper relation in the Afro-tropics (Fig. 9.3a, middle) can be explained by the saturation curve, Eqn. 9.1. It is also surprising that such a simple formulation as Eqn. 9.1 describes the pan-tropical ET characteristics because Fig. 9.3 contains the data not only from natural seasonal and rain forests but also secondary forests and plantations. From an interpretation of Budyko's vegetation categorization using the relationships among the radiation dryness index, net radiation and vegetation types (Budyko and Miller, 1974), tropical forests can exist within the ranges of annual mean net radiation of 110–133 W/m² and annual mean precipitation of 1370–5000 mm. The lowest precipitation limit for the presence of tropical forest (i.e. 1370 mm/year) is found to be broadly comparable with the lowest value of P in each ecozone (Fig. 9.3a).

The local water-use ratio (LEUR) is defined as ET divided by P and represents the ratio of P water recycling from ET . Fig. 9.3b shows these data for each tropical ecozone. The LEUR data show a significant decrease with P decreasing for all ecozones. This implies that in areas with smaller P (mainly in AFR), P is effectively recycled from ET so that the vegetation regulates its own regional water resources. In areas with

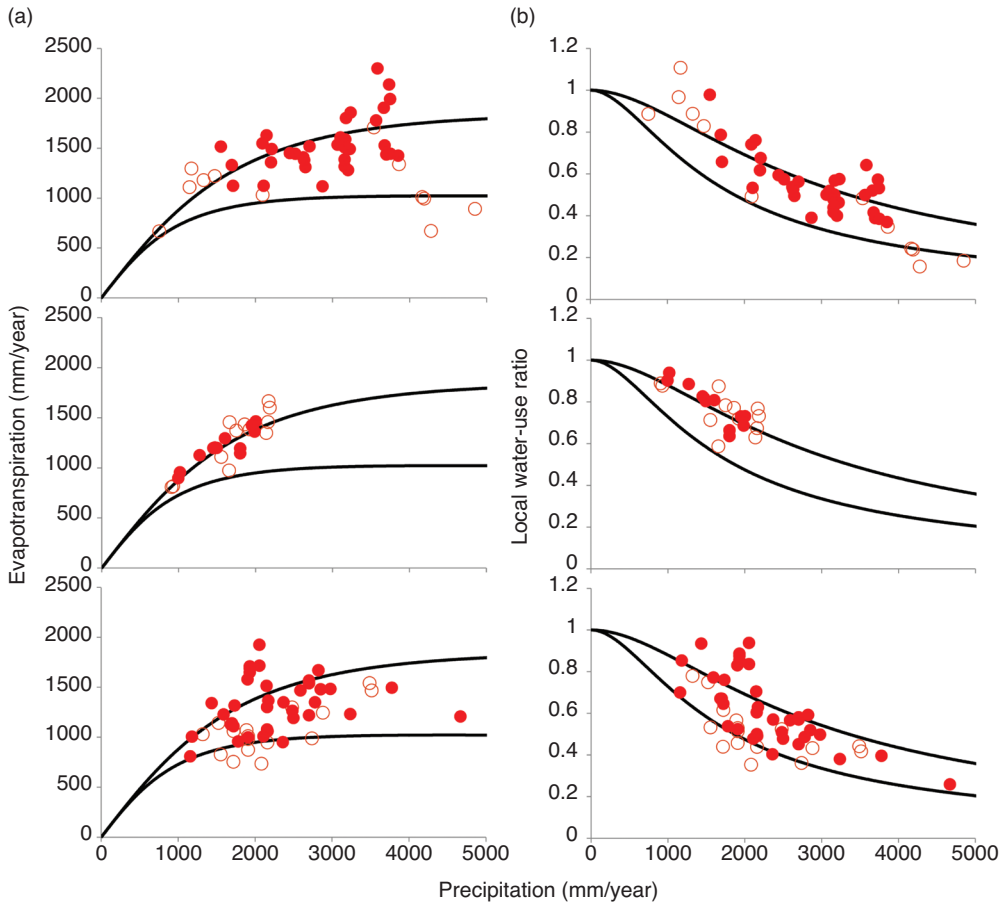


Fig. 9.3. Relationships between precipitation and (a) annual evapotranspiration and (b) local water-use ratio defined as evapotranspiration divided by precipitation, classified by the annual mean temperatures (T , °C): closed circles, $T > 25$; open circles, $20 < T < 25$. Solid lines denote the theoretical relationships: upper and bottom lines for each panel assume $T = 30$ and 15 , respectively. Top, neotropical ecozone; middle, Afrotropical ecozone; bottom, Indo–Malay–Australasia tropical ecozone. Data on evapotranspiration and precipitation taken from Komatsu *et al.* (2012).

larger P (such as NEO and IMA), P is also supplied by moist air from the surrounding terrain and oceans.

The atmosphere–land surface water balance method (Oki *et al.*, 1995) with global data on the P (Fig. 9.1) and atmospheric water vapor divergence derived from ERA-Interim gridded four-dimensional meteorology data set (Dee *et al.*, 2011) produced a pan-tropical map of ET and the LEUR. Although ET is well reproduced in areas with a relatively small amount of P , it is likely that large P in the method has induced an overestimation of ET in such areas.

Nevertheless, characteristics of spatial variation in the ET are well represented. Comparatively small ET can be seen in AFR and a much larger ET in Borneo and New Guinea Islands in the IMA tropics and Amazonian headwaters in the neotropics. Areas with lower LEUR values tend to overlap areas with larger P (compared with Fig. 9.1a), and this is consistent with the patterns shown in Fig. 9.3b. The exception is Borneo Island (IMA ecozone), where, despite a plentiful P , higher values of LEUR are estimated due to near-zero annual atmospheric water vapor divergence/convergence (Kumagai *et al.*,

2013; Kanamori *et al.*, 2018). Here, we should note that, for considering the extent to which P relies on terrestrial ET (i.e. moisture recycling), the role of global wind patterns, topography and land cover should be interpreted more in the context of continental moisture recycling (van der Ent *et al.*, 2010). Notably, Poveda *et al.* (2014) examined how 'aerial river' pathways modified by the effects of topography, orography and land-cover types contribute to precipitation patterns in tropical South America.

9.4 Pan-tropical Streamflow Generation

Fig. 9.4 shows the annual streamflow per unit drainage area (the strict definition of the term 'runoff') against latitude (determined at the confluence with the ocean; adapted from Wohl *et al.*, 2012). The streamflow for

December–February (winter in the northern hemisphere) and June–August (winter in the southern hemisphere) averaged per latitude was also estimated using the atmosphere–land surface water balance method with assumptions that deep groundwater flow contributes to streamflow at the scale of large watersheds and that changes in calculated domain water storage can be neglected (Oki *et al.*, 1995) and added to Fig. 9.4. The negative values of computed 3 month streamflow are probably attributable to the role of dynamic subsurface storage. It is apparent that the annual streamflow from tropical basins is typically much greater than that from temperate basins due primarily to the greater precipitation amounts in the humid tropics. Furthermore, the range in annual observed streamflow is much larger within the latitudes 0 to 23.4° due to the presence of extensive areas with dry climates (arid (BW) and semi-arid (BS) areas) in the tropics.

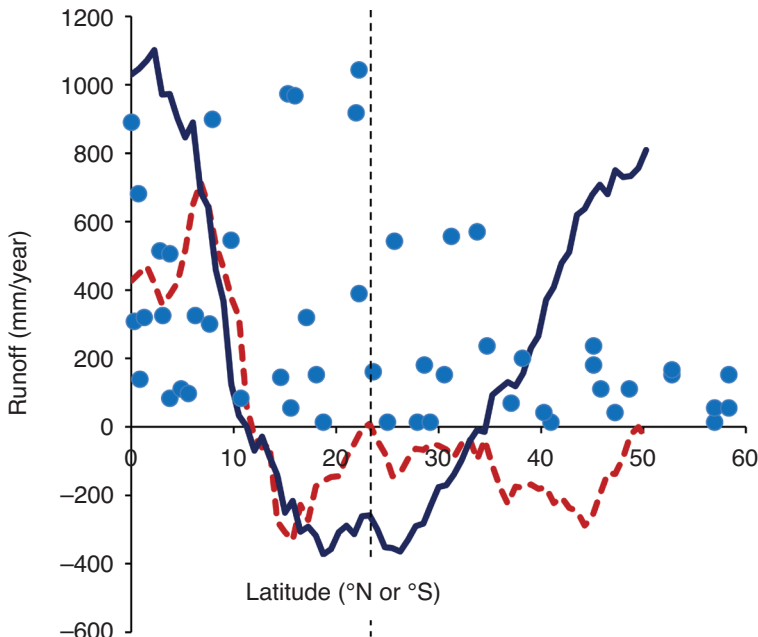


Fig. 9.4. Annually scaled streamflow per unit drainage area as a function of latitude at the river mouth (circles), adapted from Wohl *et al.* (2012). Relationships between latitude and annually scaled and latitude-averaged streamflow computed from the atmosphere–land surface water balance method for December–February in the northern hemisphere (solid line) and June–August in the southern hemisphere (dashed line) are also shown. The vertical dotted line denotes the latitude of the tropics north and south of the equator (23.4°N and S).

The greater streamflow present within the parts of the tropics with a rainforest climate and monsoon climate indicates that considerably more precipitation travels through watersheds toward streams than at other latitudes. The greater water flows within such tropical basins mean that the magnitudes of chemical and particle transport are likely to be greater (Lepcha *et al.*, 2024), and the watershed systems are more sensitive to disturbance (Wohl *et al.*, 2012). Consequently, tropical forest hydrology has implications for the other scientific disciplines of biogeochemistry and geomorphology.

Some 89% of the channel network of the globe's streams/ivers comprises first-, second- and third-order channels (see Table 2 in Downing *et al.*, 2012). This means that most streamflow is generated in the network of such low-order channels. Experimental watersheds typically comprise channels of

first- to third-order size and so are ideal locations for study of the pathways of rainwater to the channel network via surface and/or subsurface pathways. These routes of water migration to channels are known as the pathways of 'streamflow generation' or simply 'runoff pathways' (Bonell, 2004; Burt and McDonnell, 2015). Most experimental watersheds used for the study of runoff pathways are in temperate regions, with few in tropical regions (Bonell, 2004; Burt and McDonnell, 2015). Fig. 9.5a shows the locations of some key experimental watersheds with a long history of research on streamflow generation pathways in the tropics. As these are commonly located beneath tropical forests, the findings are pertinent to tropical forest environments.

Streamflow generation pathways that have been observed within these tropical forest environments include the following:

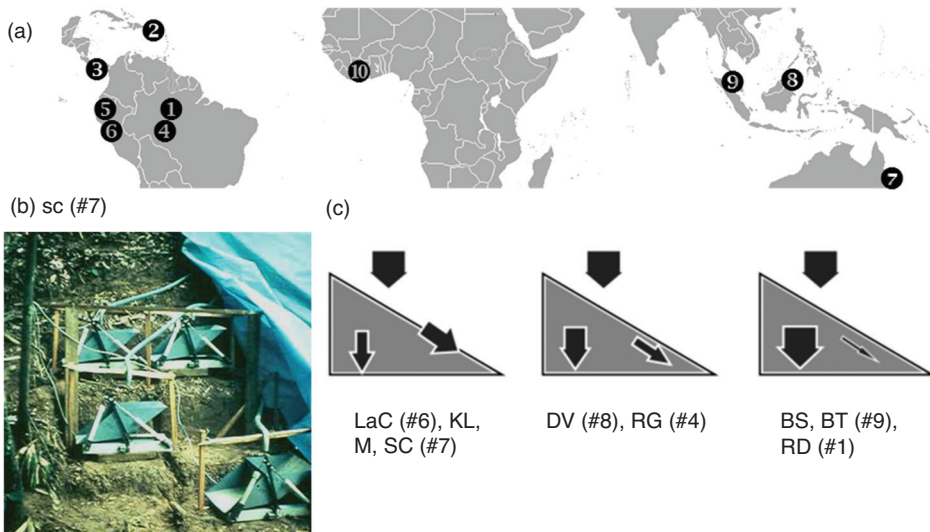


Fig. 9.5. (a) Location of ten experimental basins in the tropics with a long history of research into streamflow generation pathways: 1, Reserva Ducke (Brazil); 2, Bisley II (Puerto Rico); 3, Barro Colorado and Lutzito (Panama); 4, Rancho Grande and Jurena Ultisol (Brazil); 5, Rio San Francisco (Ecuador); 6, La Cuenca (Peru); 7, South Creek (Australia); 8, Danum Valley (Malaysia); 9, Bukit Tarek and Bukit Berembun (Malaysia); 10, Tai Forest (Ivory Coast). See Elsenbeer (2001), Bonell (2004), Hugenschmidt *et al.* (2014), and Barthold and Woods (2015) for an extensive list of publications for these and other experimental sites in the tropics. (b) The "throughflow trough" system used by Bonell and Gilmour (1978) to demonstrate the presence of shallow lateral flows in the organic and mineral soils at the South Creek Experimental Watershed (no. 7), Queensland, Australia. (c) Inferred dominant streamflow generation pathways presented in Elsenbeer (2001). Experimental watersheds: LaC: La Cuenca (#6); KL: Kiani Lestari; M: Mendolong; SC: South Creek (#7); DV: Danum Valley (#8); RG: Rancho Grande (#4); BS: Bukit Soeharto; BT: Bukit Tarek (#9); RD: Reserva Ducke (#1) (see Elsenbeer, 2001, for details).

1. *Infiltration-excess overland flow*. This water flow on slopes outside of channels is caused by precipitation falling at a rate faster than the local coefficient of permeability at the ground surface (equivalent to the 'infiltration capacity' or surface 'saturated hydraulic conductivity'; e.g. Cheng *et al.*, 2023).
2. *Saturation overland flow by direct precipitation*. Where rainfall falls on to the ground at a rate less than that of the infiltration capacity but where pores are already saturated, then no further infiltration can take place, and new water travels over the surface.
3. *Subsurface flow*. Where rainfall infiltrates, some will evaporate from the soil or support transpiration from vegetation, and the remainder will travel toward streams below the surface. Most of this water will enter the streams via the channel bed and banks, but some will return to the surface prior to reaching a channel (so-called return flow) and flow over the surface as saturation overland flow. Flow beneath the surface may be very shallow where lithomorphous soils overlie impermeable geology but may be over 100m deep where permeable soils overlie permeable geology (whether unconsolidated materials or rock).

Some ambiguity in the definition of water pathways arises from the definition of the ground surface. Some scientists define overland flow as water moving over the surface of a mineral A (or E) soil horizon, while others use the definition of water moving laterally above the typically overlying organic horizons (i.e. L, H and/or O horizons).

Streamflow generation studies in the tropics that have directly observed the presence of overland flow where the rainfall intensity exceeds the infiltration capacity are very limited, but the Tai Forest basin (Bonell, 2004) and Basper basin (Cheng *et al.*, 2023) are good examples. Studies that have observed overland flow on permeable but saturated soils include the South Creek (Bonell and Gilmour, 1978) and La Cuenca (Elsenbeer and Vertessy, 2000) basins. Several studies have direct evidence of lateral (downslope) flows within the subsurface generated during rainstorms. Bonell and Gilmour (1978) observed the presence of these flows using so-called 'throughflow troughs'

(Fig. 9.5b), while Chappell and Sherlock (2005) observed the presence of these flows by monitoring tracer migration.

The more pertinent question is not whether a particular pathway is present but whether it is the dominant pathway producing more than 50% of the observed total streamflow, particularly during storm events (i.e. without any manipulation of hydrographs by separation methods). Too often, researchers infer the dominance of a particular pathway simply from the observed presence of the pathway, rather than relating measured flows per unit basin area from that pathway to those observed in the stream per unit basin area. For example, the measured lateral flows per unit basin area of Gilmour *et al.* (1980) are only a tiny fraction of the observed streamflow per unit basin area. Consequently, the importance of measured near-surface flows at this site (and many other sites) has been over-emphasized in comparison with the unmeasured flows within deeper soil and unconsolidated rock strata. The resultant ambiguity and misinterpretation of the dominant pathway is amply illustrated by inconsistencies in the inferred dominant pathways presented in the reviews of Elsenbeer (2001), Bonell (2004) and Barthold and Woods (2015), as shown in Fig. 9.5c and Table 9.1.

A further area of concern is the focus of most studies on pathways in the solum (i.e. A and B soil horizons) alone, as highlighted by Bonell and Balek (1993) and Bonell (2004). There is an increasing awareness that at some tropical sites, soils may be developed on unconsolidated geological materials that are permeable and have deeper pathways within these strata. Examples of experimental watersheds developed on these deeper porous media in the tropics include the Lake Calado microbasin near the Reserva Ducke basin in Brazil (Lesack, 1993), the Jungle Falls basin in Singapore (Chappell and Sherlock, 2005; Rahardjo *et al.*, 2010), the Bukit Timah (Noguchi *et al.*, 2005) and Bukit Berembun (Chappell *et al.*, 2004) basins in Malaysia, and the O Thom II basin in Cambodia (Shimizu *et al.*, 2007). Equally, other low-order basins in the tropics may be located on rock aquifers or rocks with fracture systems that produce very deep water pathways. The Arboleda basin near La Selva, Costa Rica (Genereux *et al.*, 2005) and the Jatunhuaycu hillslope in the tropical Andes

Table 9.1. Inferred dominant streamflow generation pathways presented in Table 14.1 of Bonell (2004). See Bonell (2004) for further details of the experimental watersheds and pathway definitions.

Dominant pathway	Experimental watershed
Predominantly vertical pathways	Reserva Ducke (#1), Bukit Tarek (#9), Fazenda Dimona, Mgera
Predominantly lateral pathways (infiltration-excess overland flow)	Tai Forest (#10), Basper (Cheng <i>et al.</i> , 2023)
Predominantly lateral pathways (saturation-excess overland flow)	South Creek (#7), Barro Colorado (#3), La Cuenca (#6), Maburao
Predominantly lateral pathways (subsurface stormflow within the soil)	Danum Valley (Sungai Baru Barat) (#8), Rancho Grande (#4), Bisley II (#2), Dodmane, Kannike, Ife, ECEREX, Sekancing (Sayama <i>et al.</i> , 2021), Agua Salud (Bush <i>et al.</i> , 2020)
Predominantly lateral pathways (subsurface stormflow at the soil-bedrock Interface)	Kuala Belalong

each # represents the location in Fig. 9.5a.

(Lahuatte *et al.*, 2022) have such pathways. To alert researchers to the potential presence and role of these deeper streamflow generation pathways, Chappell *et al.* (2007) developed a perceptual model of runoff pathways where a type III system has a dominance of pathways via unconsolidated geological materials, and a type IV system has a dominance of pathways via rock aquifers or fracture systems (Fig. 9.6). Experimental basins lacking any evidence of major pathways through geological strata (solid rock or unconsolidated materials) are classified as either type II systems (where a B soil horizon is present, e.g. Chappell *et al.*, 1998) or type I systems (where a lithomorph soil is developed on steep, impermeable mountain slopes). The potential presence of deeper pathways within existing experimental watersheds needs to be a key focus for new research. With complete observational evidence, including information gained from both hydrometric and tracer studies (as recommended by Barthold and Woods, 2015), researchers may be closer to developing a unified, numerical model of the dominant pathways of streamflow generation applicable across all tropical forest environments.

9.5 Research Needs

The projected growth in atmospheric greenhouse gases within the coming century, as predicted by the Intergovernmental Panel on

Climate Change’s (IPCC) A1B (balanced across all sources) scenario, will significantly increase tropical surface temperatures ranging from ~3°C to 5°C for South-east Asia, the Amazon and West Africa (IPCC, 2007). In addition, the A1B scenario predicts concomitant modifications to precipitation patterns: a general increase in precipitation for West Africa, intensification of the seasonality of precipitation for South-east Asia (i.e. more and less precipitation in the wet and dry seasons, respectively) and a general decrease in precipitation for the southern and eastern Amazon. More dramatically, as an example, the 1997–1998 El Niño was the strongest in the 20th century, and its associated drought in Borneo Island in South-east Asia was the most severe (statistically, a drought such as in 1998 may occur once in ~360 years). Tree mortality rates during that drought were 6.37%/year, as compared with 0.89%/year during the pre-drought period (1993–1997) in the studied forest site in western Borneo (Nakagawa *et al.*, 2000). Climate models project an increase in the frequency and intensity of ENSO activity (El Niño and La Niña events) in response to global warming (Capotondi, 2015). This will likely also induce severe droughts and devastating floods in the South-east Asian tropics (Capotondi, 2015). Again, tropical forest regions are crucial for climate mitigation through carbon sequestration and evapotranspiration. In this context, Takamura *et al.* (2023) characterized the relationship between the ENSO-induced

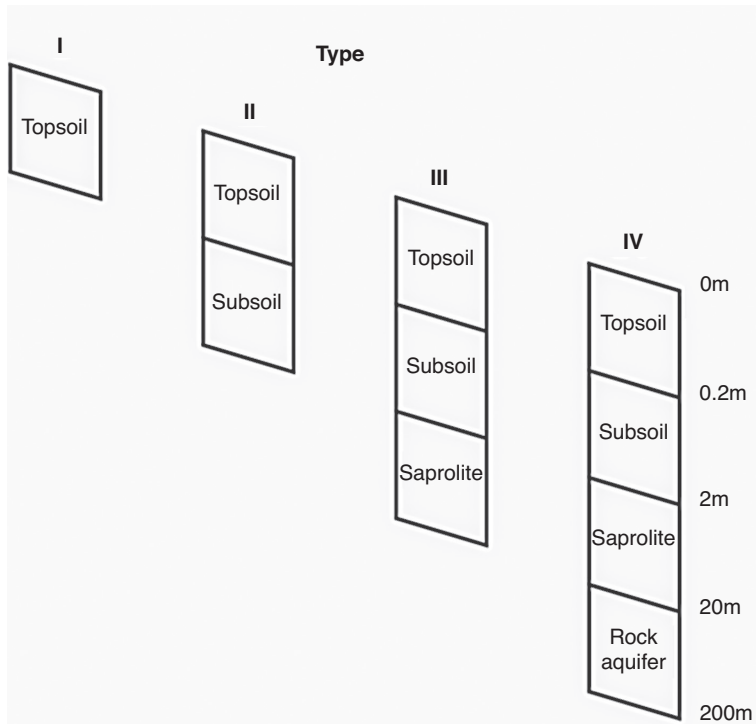


Fig. 9.6. Schematic representation of water pathway systems within tropical forests classified according to the presence of as many as four strata (A soil horizon or topsoil, B soil horizon or subsoil, unconsolidated geological materials and solid rock) named types I to IV. Example depths and the logarithmic depth scale are given. Modified from Chappell *et al.* (2007) and US Department of Energy (2012).

climatic conditions and atmosphere–forest canopy exchange in a South-east Asian rainforest, suggesting the long-term climatic influence on the carbon and water cycles and ecosystem regulation mechanisms in tropical rainforests.

In addition, the tropics are known to be a very active domain in terms of changes in land cover. According to an investigation by Hansen *et al.* (2013) using Earth observation satellite data from 2000 to 2012, the tropics were the only domain to exhibit a statistically significant increasing trend in annual forest loss. This was 2101 km²/year and tropical rainforests loss was 32% of the global forest loss. South American tropical dry forests have been lost at the highest rate in the lost tropical forests. Furthermore, Hansen *et al.* (2013) confirmed that the tropical forest loss rate in Brazil has decreased from 40,000 km² in 2003–2004 to 20,000 km² in 2010–2011, while forest loss in

Indonesia drastically increased from 8000 km² in 2000–2001 to 20,000 km² in 2010–2011, indicating that an increase in Asian tropical forest loss ‘compensated’ for a decrease in Amazonian tropical forest loss. Despite a drastic reduction in the deforestation rate in the whole world in 2015–2020 compared with the rate in 1990–2010, deforestation and land-use change in the tropical domain are still ongoing, especially in the African tropics. The tropical deforestation intensity is in the order Africa >> South America > South east Asia (FAO, 2020).

In the humid tropics, there is plenty of precipitation, and a combination of climate change and drastic changes in land covers must induce pressure for freshwater resources and loss of soil via changes in local hydrological processes. Tropical forest evapotranspiration is generally greater than evapotranspiration from grasslands. Hence changing land covers,

such as from forests to pastures, would reduce evapotranspiration and increase streamflow, resulting in, in some cases, increasing flood frequencies (Bruijnzeel, 2001, 2004). The generation of localized infiltration-excess overland flow on compacted soil surfaces caused by forestry operations accelerates the geomorphological process of soil erosion (Bruijnzeel, 2004; Sidle *et al.*, 2006; Sidle and Ziegler, 2012; Douglas, 2022). Such land degradations would become increasingly worse by altered precipitation regimes through climate change (Peña-Arancibia *et al.*, 2010; Wohl *et al.*, 2012). However, the hydrological properties of secondary vegetation, such as evapotranspiration and surface infiltration, may quickly resemble those of the original forest again (Giambelluca, 2002; Bruijnzeel, 2004).

From a global perspective, it should be emphasized that tropical forests are a major source of global hydrological fluxes. Changes in evapotranspiration rates could significantly impact both global and regional climates, in turn potentially affecting feedback to the atmosphere (Bonan, 2008). Many previous research contributions to the knowledge of hydro-climate in the tropics have suggested that: (i) forest cutting has more effect on local and global climate under maritime conditions than under continental conditions (van der Molen *et al.*, 2006; Takahashi *et al.*, 2017; Kanamori *et al.*, 2018); (ii) rainwater is recycled earlier by wet canopy evaporation than via transpiration (van der Ent *et al.*, 2014); and (iii) in the maritime continent, changes in sea surface temperature influence the precipitation regimes more than the land-cover changes (Bruijnzeel, 2004). Furthermore, large-scale tropical deforestation and selective

logging could result in warmer and drier conditions at the local scale, and also the teleconnections from the converted tropical lands could pose a considerable risk to agriculture in other regions due to impacts on precipitation against a background of warmer temperatures (Lawrence and Vandecar, 2015).

As the first step for considering the future hydrological impact in the tropics, we need an understanding of the current and basic tropical hydrological cycling, as well as the hydrological interactions among the earth's surface and subsurface, vegetation and atmosphere, based on long-term and networked data acquisition and organization. According to Cui *et al.* (2022), who investigated water resource availability altered by global vegetation changes, we can determine how much the local terrestrial water availability has decreased with the reduction of vegetation covers in the tropical forest regions. As Wohl *et al.* (2012) pointed out, field-based hydrological measurements in many tropical countries have been less explored than those in temperate regions, and to make matters worse, are waning. We should note that the lack of long-term observations throughout the tropics leads to a failure to validate hydro-climate models and, thus, an impossibility for extrapolating future tropical forest hydrology using the output of the models, which must be built with reference to reliable observations and should use these observations as inputs (e.g. Duvert *et al.*, 2022). Besides making more effort to organize a field-based hydrological observation network over the tropics, as shown in this chapter, remote sensing technologies and global-scale climate and geoinformation databases can serve among the most promising tools to complement the lack of tropical field-observation sites.

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